



The Application Of C++ For A Better Water Intake

Group 3 - Programming



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OUR TOPICS

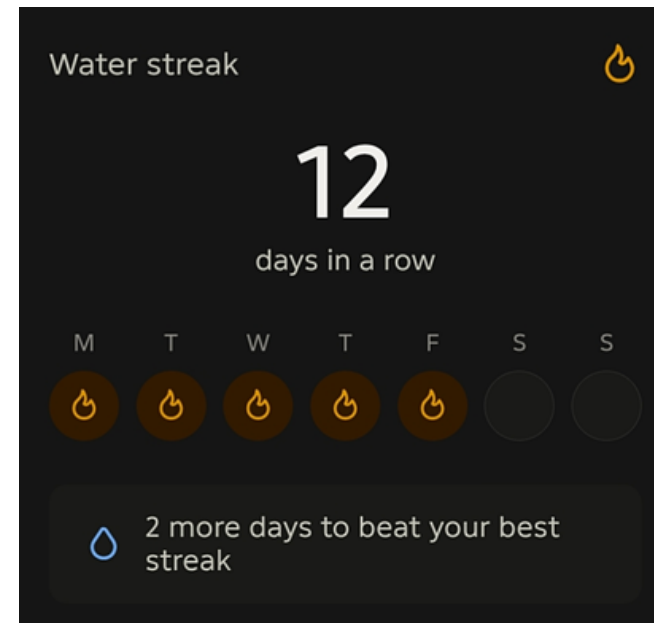
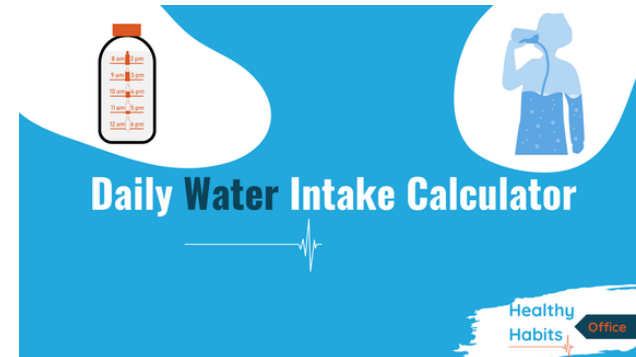
- Problems
- Solutions
- Features
- Advantages and Disadvantages
- Flowchart
- Conclusion

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Problems



Solutions



Features

Water Intake Calculator

Calculates the water intake needed and creates a target depending on weight, climate, and activity.

Water Intake Reminder

Gives notification to drink water to fulfill the water intake target.

Water Streak

For extra excitement and motivation to drink water.

Advantages/Disadvantages of the app

Advantages:

- Helps keep track of daily water intake and provides insight on individual's water level
- Builds habit through consistency by following the app's reminder
- Prevents dehydration which could cause fatigue

Disadvantages

- If the individual does not follow the app properly, could cause dehydration
- Water intake in an individual could be inaccurate as water from foods are not accounted

Advantages/Disadvantages of C++

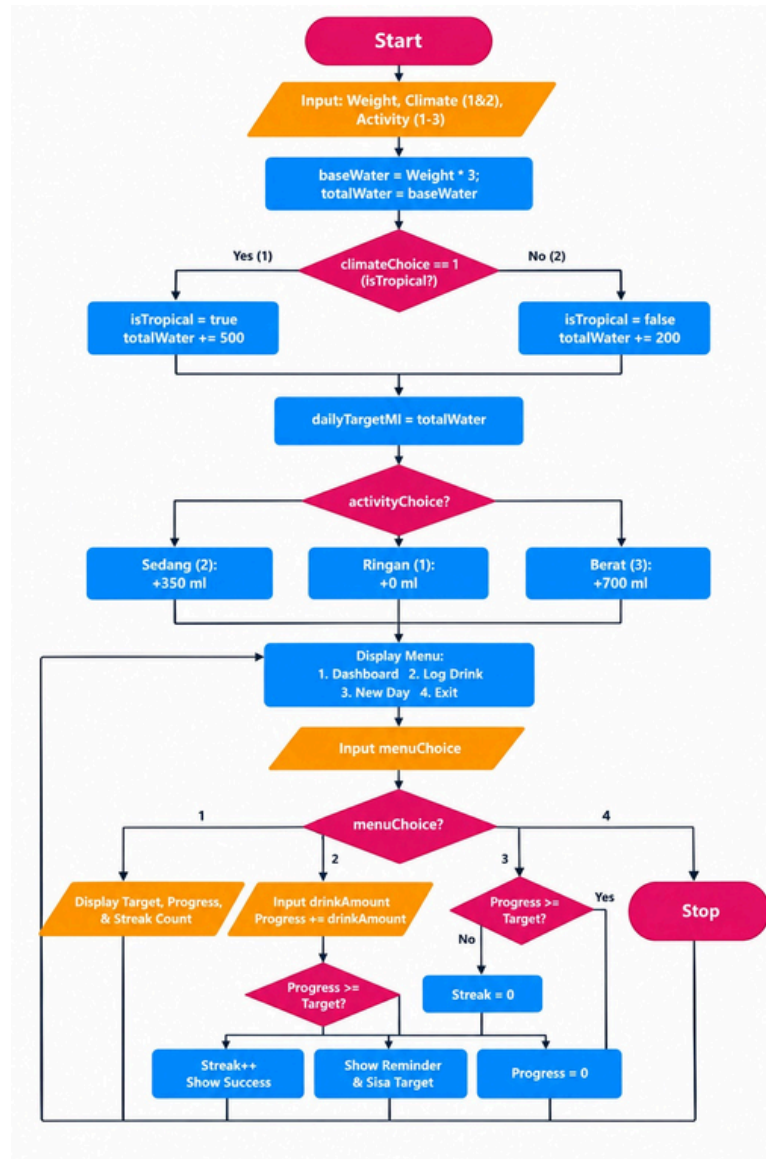
Advantages:

- C++ allows the app to be used across platforms
- Through C++ app could be accessible for offline users
- Constant notifications to remind the users of water intake

Disadvantages

- Hard to setup built-in UI for mobile users through C++

Flowchart



Conclusion

The Problem: Dehydration quickly leads to fatigue, headaches, and poor concentration

The Solution: A C++ app featuring a personalized offline calculator to set daily targets, alongside offline reminders and "Water Streaks" to build consistent habits.

The Trade-offs: C++ enables cross-platform, offline access, but building a mobile UI is difficult, and its effectiveness depends entirely on user discipline.

References:

- Armstrong, L. E., Ganio, M. S., Casa, D. J., Lee, E. C., McDermott, B. P., Klau, J. F., Jimenez, L., Le Bellego, L., Chevillotte, E., & Lieberman, H. R. (2012). Mild dehydration affects mood in healthy young women. *The Journal of Nutrition*, 142(2), 382–388. <https://doi.org/10.3945/jn.111.142000>
- Ganio, M. S., Armstrong, L. E., Casa, D. J., McDermott, B. P., Lee, E. C., Yamamoto, L. M., Marzano, S., Lopez, R. M., Jimenez, L., Le Bellego, L., Chevillotte, E., & Lieberman, H. R. (2011). Mild dehydration impairs cognitive performance and mood of men. *British Journal of Nutrition*, 106(10), 1535–1543. <https://doi.org/10.1017/S0007114511002005>



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